Report_DEB

1. Importation des données

```
file_path <- here::here("mod/C_DEB_SAFLYe")

l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
  )

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)
```

2. Model fit

```r
NbIter <- 100000
seeds <- c(C1 = 3443, C2 = 6546, C3 = 111112)

text_priors <- '
Distrib (Fm,           TruncNormal_cv,          5,      0.2,  0.01,      10);
Distrib (pAm,          TruncNormal_cv,        100,      0.2,    40,    2000);
Distrib (r_pAm_pM,     TruncNormal_cv,        0.8,      0.1,    0.1,      2);
Distrib (v,            TruncNormal_cv,    0.18505,      0.1,  0.00001,    2);
Distrib (kap,          Uniform,                0.01,     0.99);
Distrib (Ehp,          TruncNormal_cv,         100,      0.2,    40,    300);
Distrib (E_coc,        TruncNormal_cv,          30,      0.2,     1,    300);
Distrib (rOM_ClxHorse, LogUniform,         0.00001,      0.3);
Distrib (dv,           TruncNormal_cv,           2,      0.5,     0,     10);
Distrib (wE,           LogUniform,      0.000000001,     0.1);

# Calcul des vraissemblances
Distrib (Sigma_W, TruncNormal,  5, 1, 0.01, 100);'

text_likelihood <- 'Likelihood (Weight,       Normal, Prediction(Weight), Sigma_W);
Likelihood (Reproduction, Poisson, Prediction(Reproduction));'

OM_diff <- TRUE
RTOL <- 1e-6
ATOL <- 1e-6

# makemcsim DEBcali.model

f_In_tot(file_path, l_Experiments, Adults_alone, OM_diff, text_priors, text_likelihood, NbIte

# ./mcsim.DEBcali DEB_C1.in
```

`OM_horse` and `OM_soil` distinct ? TRUE

Adult and alone earthworm taken into account ? FALSE

3. Results

```
                 Fm.1.    pAm.1. r_pAm_pM.1.       v.1.      kap.1.    Ehp.1.
Result_mode 1.6642200 181.44300  0.78923400 0.19319300 0.89503400 114.72000
2.5%        1.4705600 150.10500  0.67902700 0.15304350 0.85074900  79.41200
97.5%       2.3659100 191.13700  0.90323450 0.22598600 0.91920800 137.43500
```

```
Min         1.2205200  85.65750  0.61771300 0.12654700 0.30975300  64.56610
Max         6.0035700 203.19300  0.98699300 0.26973900 0.93002300 157.11000
Result_mean 1.8874863 171.53630  0.79377449 0.18772386 0.88799062 109.15600
Result_sd   0.2341197  10.38218  0.05669675 0.01917534 0.01718977  15.02521
              E_coc.1. rOM_ClxHorse.1.    dv.1.         wE.1. Sigma_W.1.
Result_mode  4.3025700    0.0013753000 4.375090 9.827170e-06 0.11241500
2.5%         2.8164710    0.0006260844 3.914710 6.752010e-09 0.10075000
97.5%        5.2762100    0.0026232900 5.872620 1.261080e-05 0.12739100
Min          2.2994500    0.0001494400 1.053680 1.008130e-09 0.09064190
Max         35.2154000    0.2742750000 6.702290 1.543700e-05 5.93830000
Result_mean  4.0288380    0.0014652236 4.809692 8.378787e-06 0.11392429
Result_sd    0.6810572    0.0016738209 0.506979 3.459369e-06 0.04849207

Number of observations, n =  277
Number of parameters, k =  10
Log-likelihood, LL =  5.355922
BIC =  45.52833


Potential scale reduction factors:

                Point est. Upper C.I.
dv.1.                 1.18       1.54
E_coc.1.              1.04       1.10
Ehp.1.                1.09       1.28
Fm.1.                 1.06       1.18
kap.1.                1.23       1.65
pAm.1.                1.36       1.98
r_pAm_pM.1.           1.07       1.21
rOM_ClxHorse.1.       1.01       1.02
Sigma_W.1.            1.02       1.07
v.1.                  1.00       1.01
wE.1.                 1.58       4.33

Multivariate psrf

1.49
```

```
mcmc_list_corr <- mcmc.list(
  lapply(mcmc_list, function(x) {
    mcmc(as.matrix(x)[-1, , drop = FALSE])  # <- reconvertir en mcmc
```

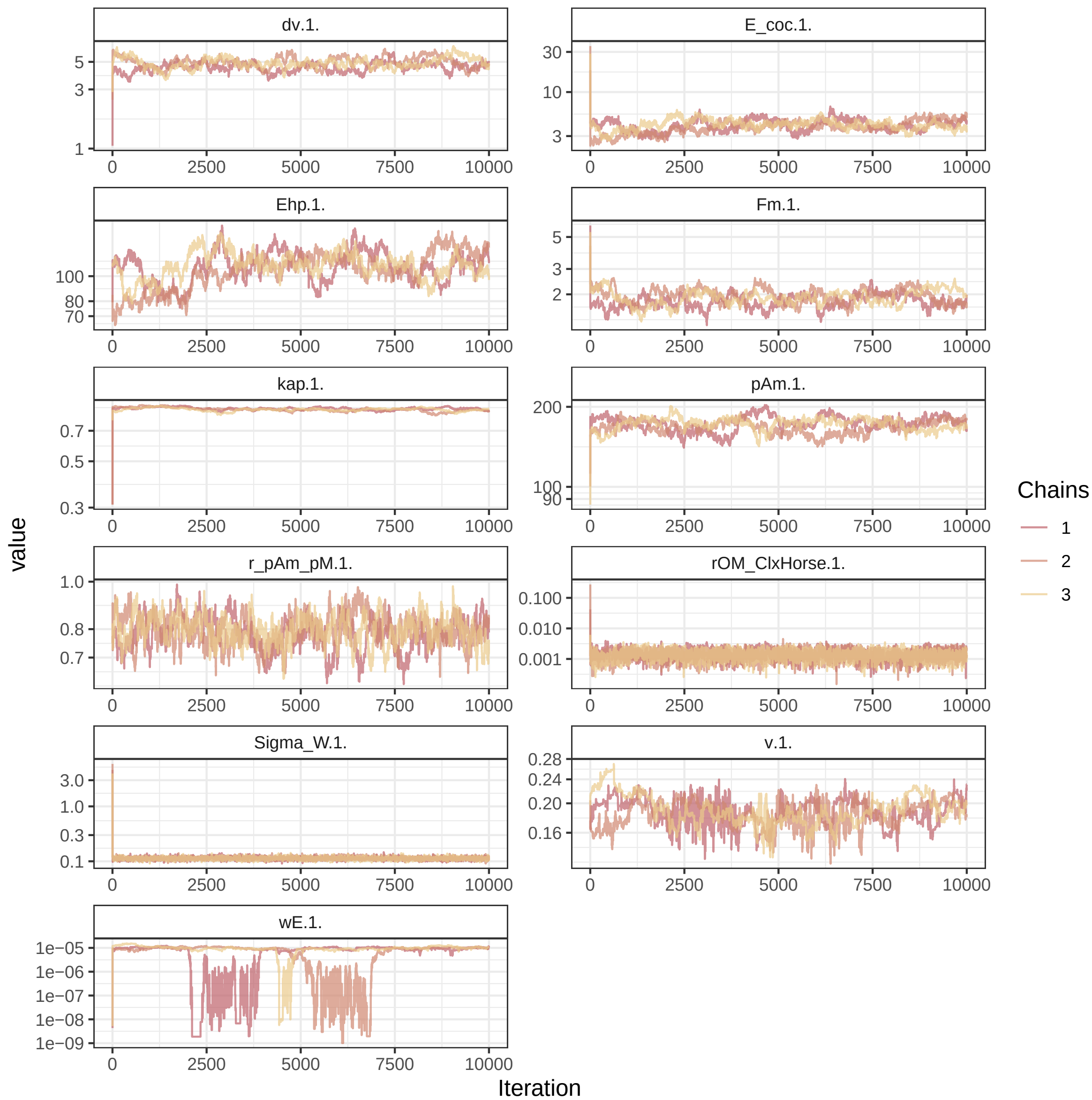


Figure 1: Parameters chains. Only the first iteration and the last `r Nb_iter_kept` are represented.

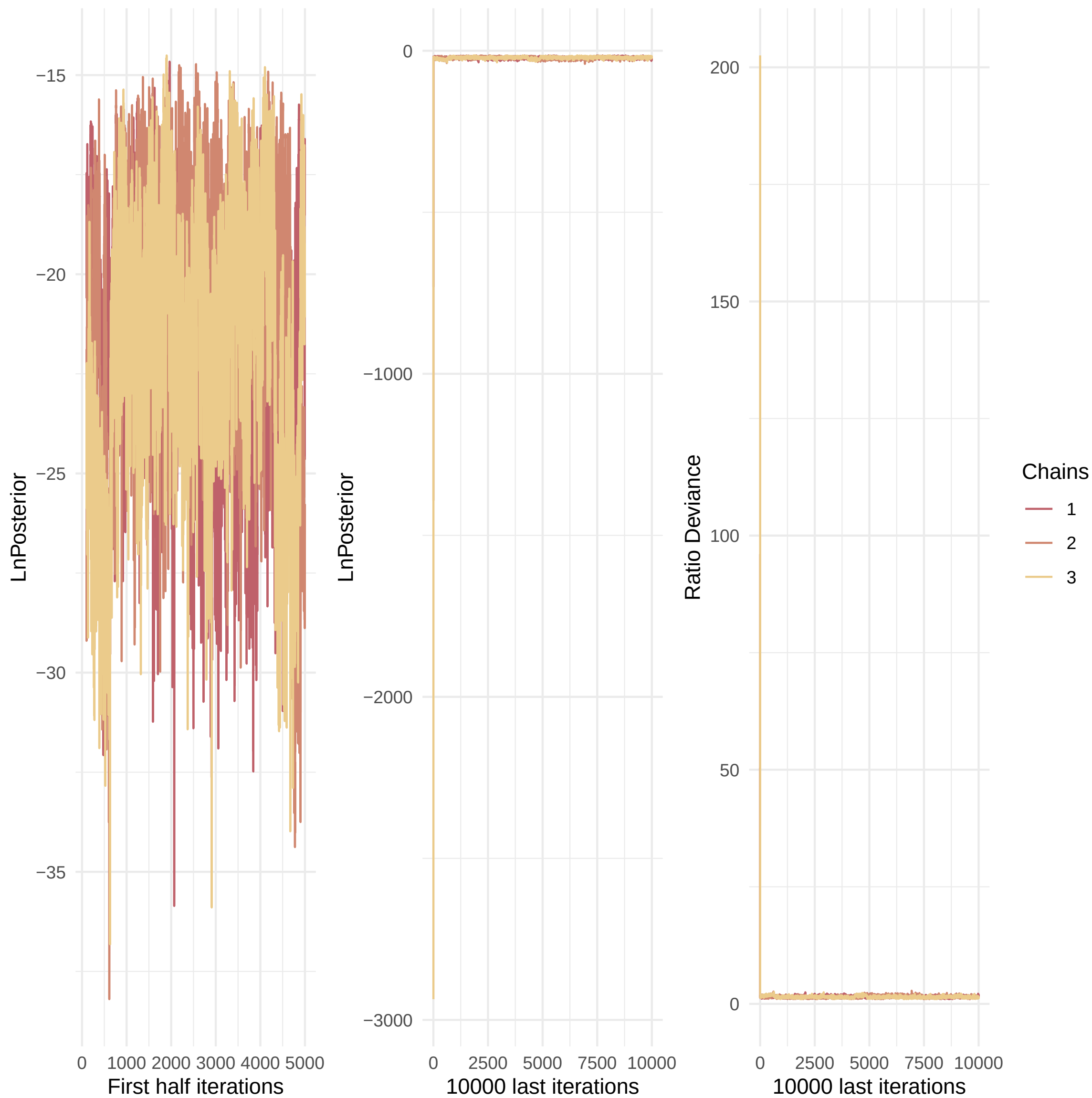


Figure 2: Evolution of likelihood and deviance ratio through the iterations.

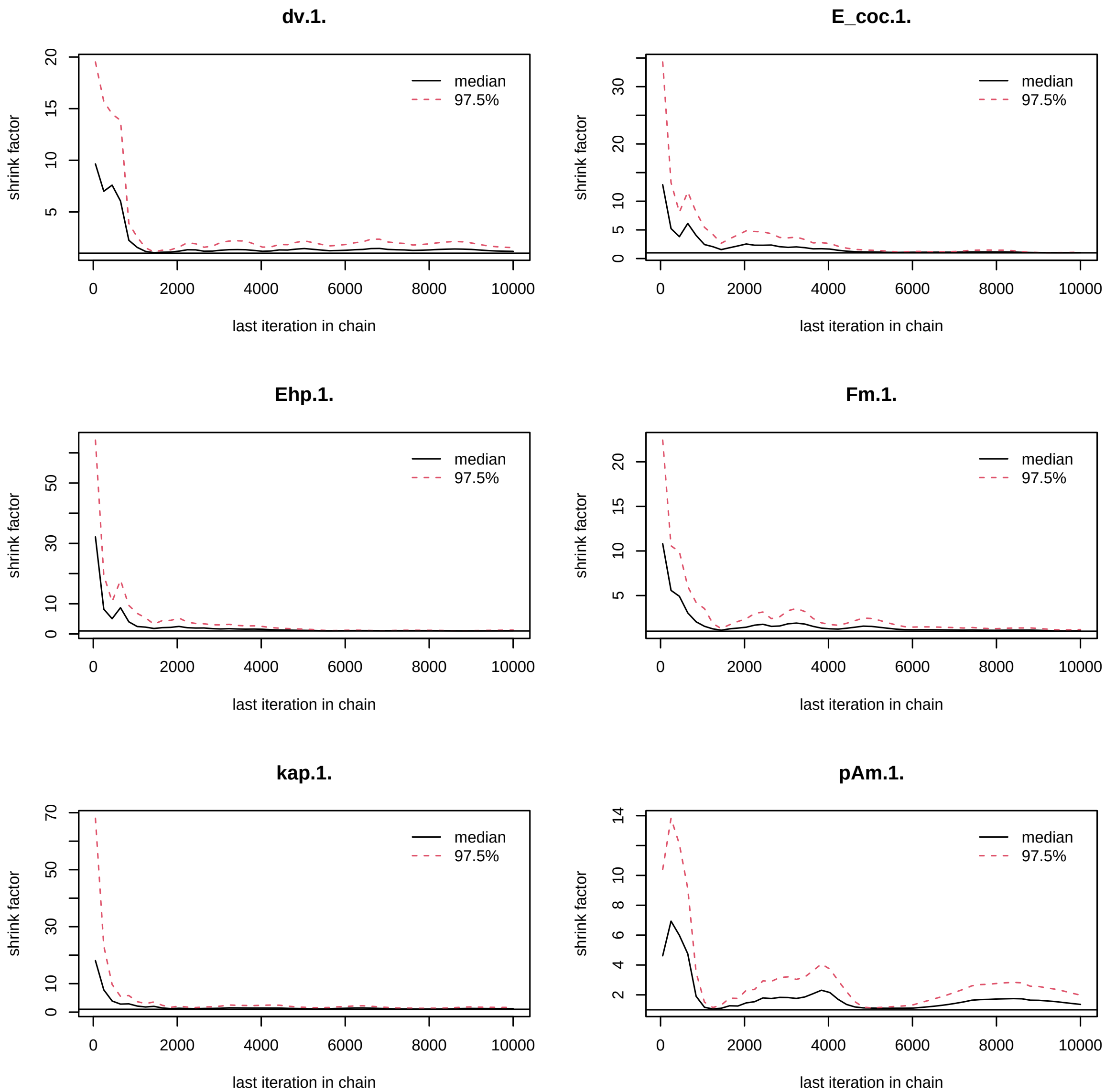


Figure 3: Chains convergence.

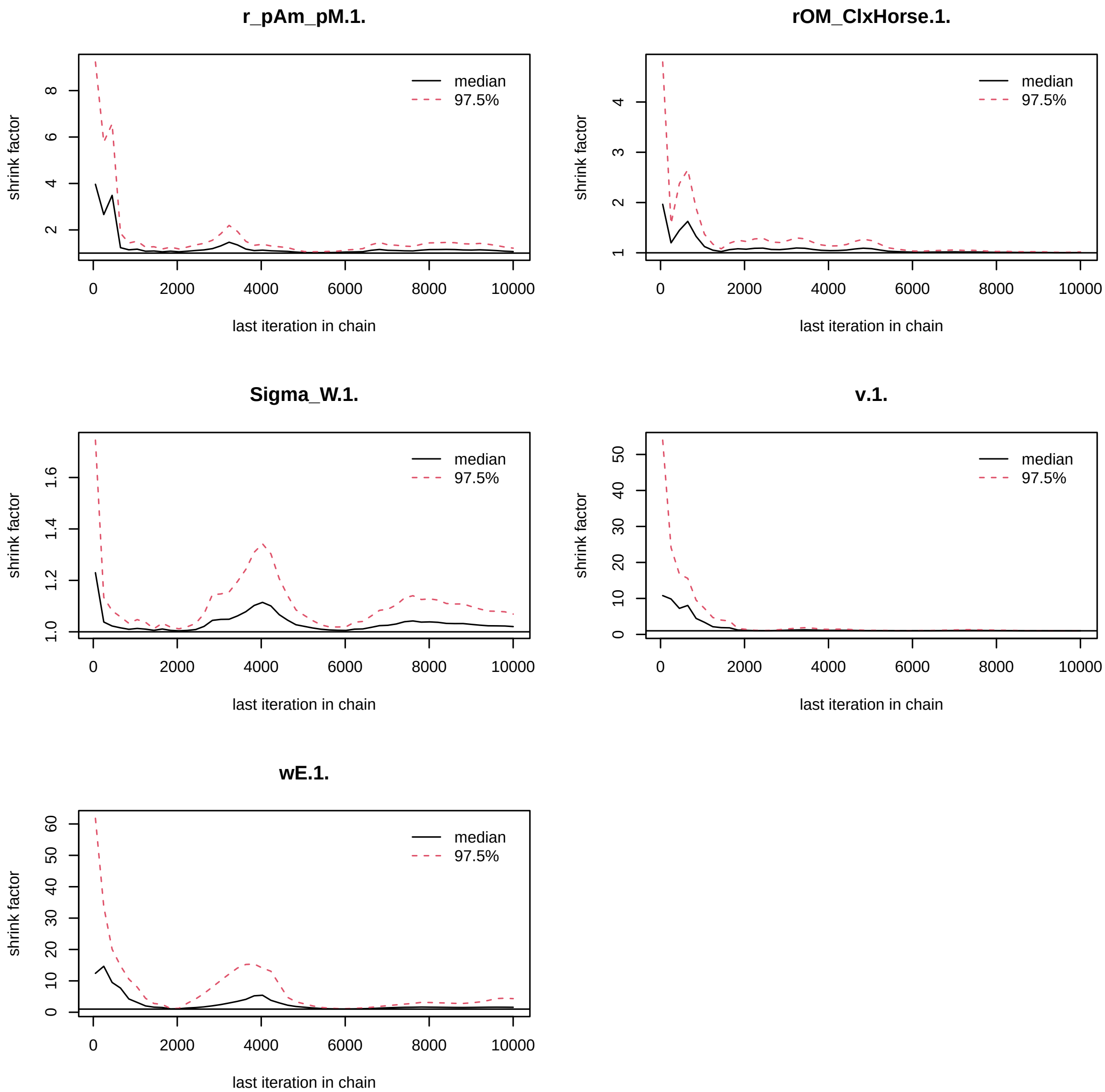


Figure 4: Chains convergence.

```
  })
)

colramp_aurora <- colorRampPalette(rev(Nord_aurora[1:4]))
ipairs(as.matrix(mcmc_list_corr), colramp = colramp_aurora, pixs=0.25, cex.diag = 0.5)
```

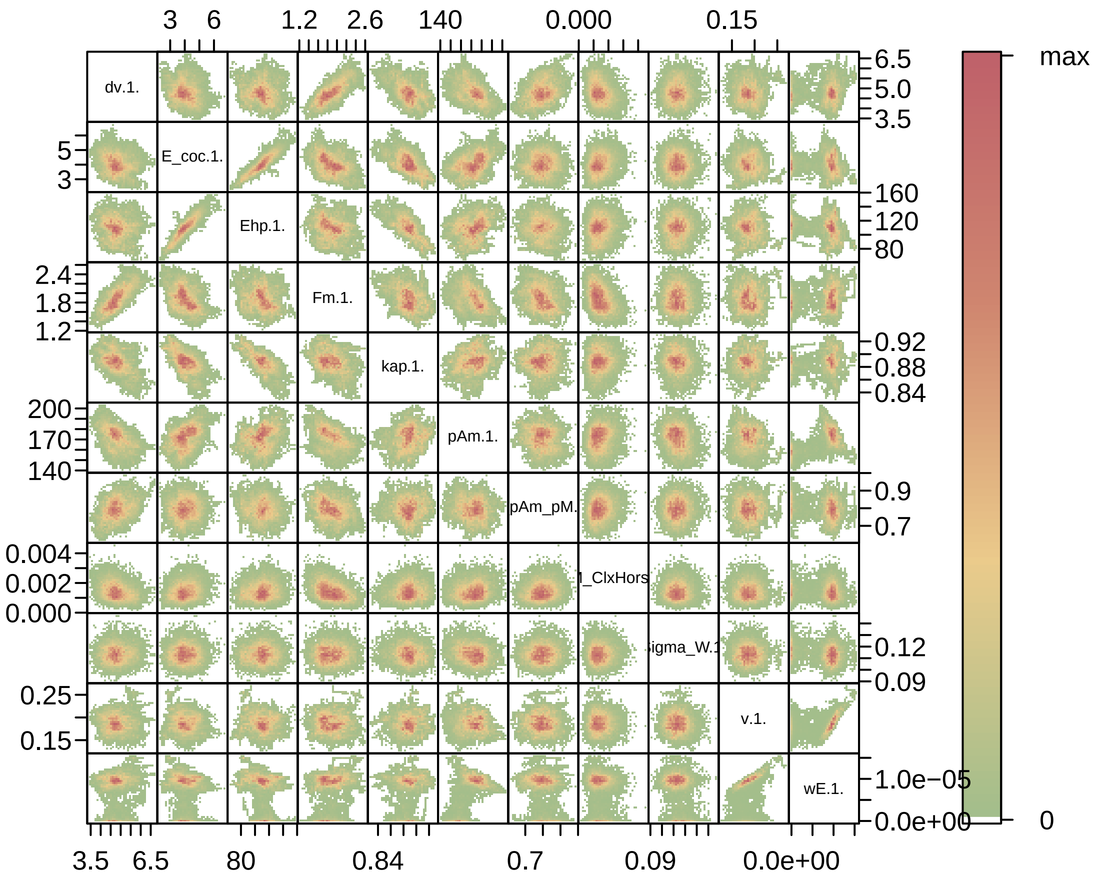


Figure 5: Parameter correlations.

4. Simulations

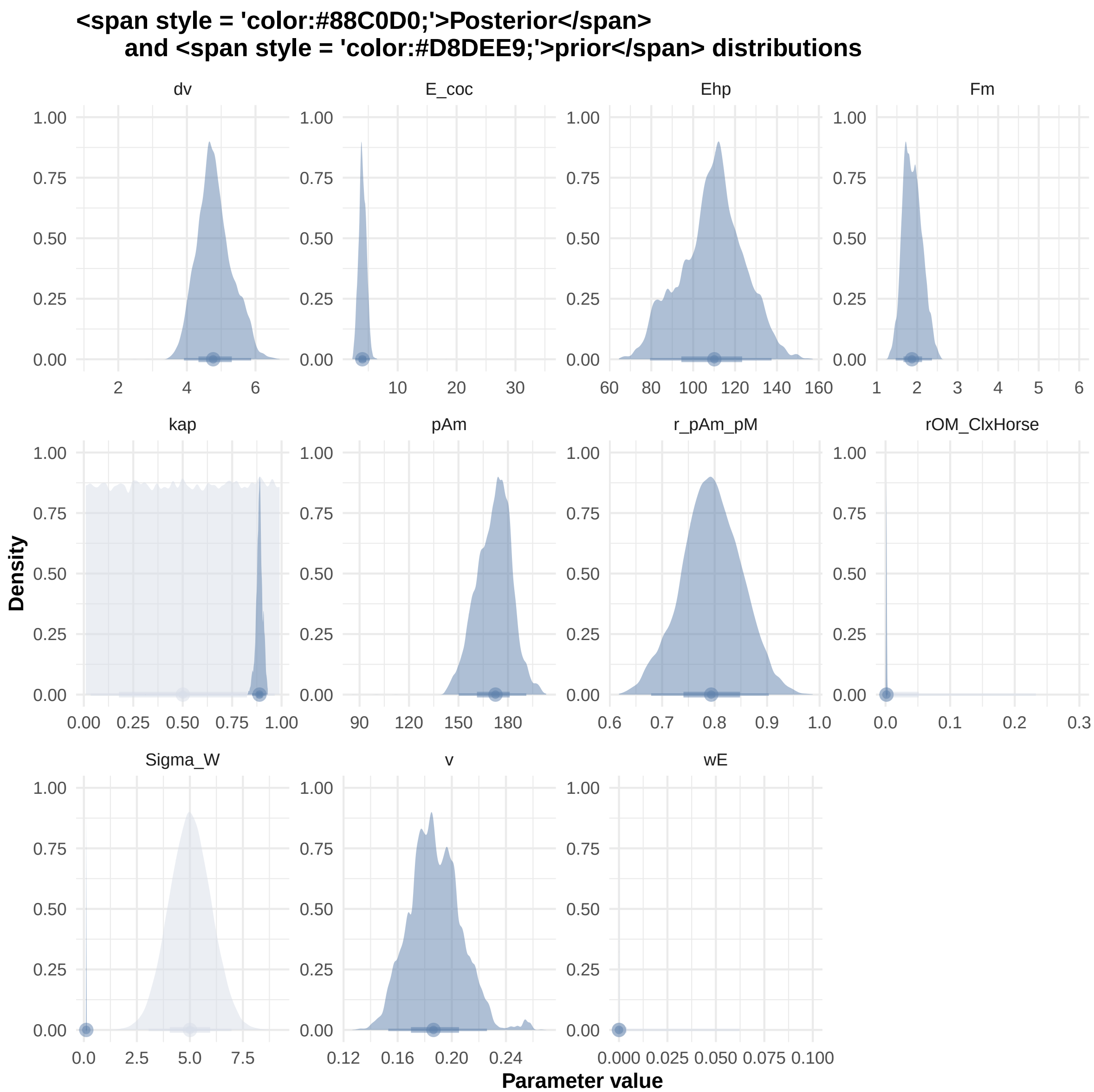


Figure 6: Distributions of the priors and the posteriors.

```r
l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
  )

Adults_alone <- TRUE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)

text_param <- Summary_res["Result_mode", , drop = FALSE] %>%
  t() %>%
  as.data.frame() %>%
  tibble::rownames_to_column("param") %>%
  rename(val = 2) %>%
  mutate(
    param = gsub("\\.1\\.$|\\.1\\.", "", param),
    line  = glue("{param} = {val};")
  ) %>%
  pull(line) %>%
  paste(collapse = "\n")

Endpoints_print <- "Weight,
    Length_struct,
    Reproduction,
    Energy,
    Organic_matter,"

f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff, text_param, Endpoints_prin
#f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff=FALSE, text_param, text_pr
```

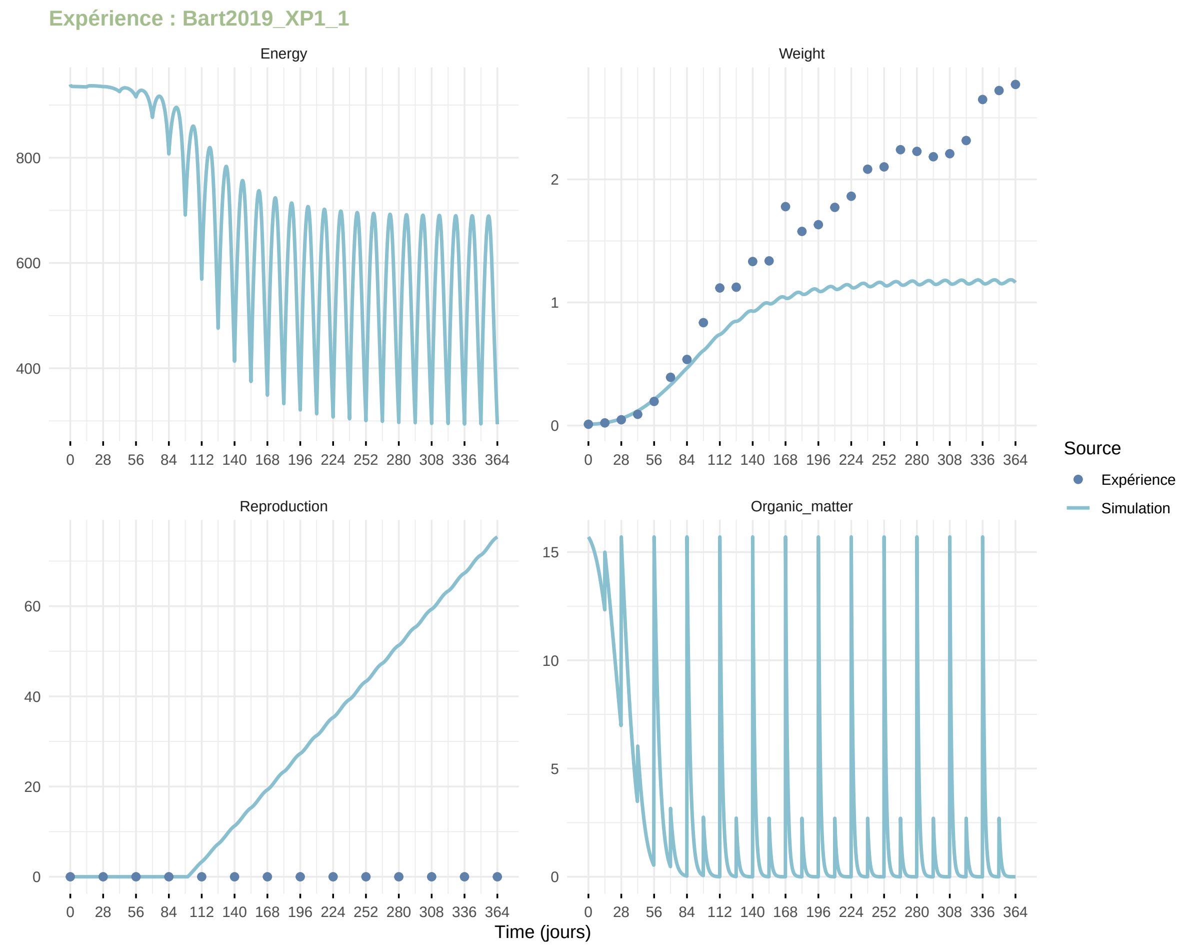
```
# ./mcsim.DEBcali DEB_sim_full.in
```

```
##############################
df_No_sim <- df_data |>
  dplyr::select(ID_experiment, No_sim)

df_carac_color <- data.frame(
  carac = unique(l_carac),
  color = c(Nord_aurora[1], Nord_polar[2], Nord_aurora[3], Nord_aurora[2], Nord_aurora[5],
            Nord_aurora[5], Nord_frost[4], Nord_frost[3], Nord_frost[2], Nord_frost[1])
)
v_color_carac <- setNames(df_carac_color$color, df_carac_color$carac)
#############################

df_data <- f_import_data_DEB(l_Experiments, Adults_alone) |>
  left_join(df_carc_exp, by="ID_experiment")

# time_limits <- df_data %>%
#   group_by(ID_experiment) %>%
#   summarise(time_cutoff = max(Time, na.rm = TRUE) + 5)

select_times <- seq(0,400, 0.1)

df_sim <- f_MCSim_read_sim(here::here(file_path, "sim.out")) |>
  left_join(df_No_sim, by="No_sim") |>
  left_join(df_carc_exp, by="ID_experiment") |>
  # left_join(time_limits, by = "ID_experiment") |>
  # filter(Time <= time_cutoff) |>
  # dplyr::select(-time_cutoff) %>%
  filter(Time %in% select_times) |>
  mutate(
    Reproduction = case_when(
      Reproduction <= 5e-6 ~ 0,
      .default = Reproduction
    )
  )
```

```
Warning: `read_table2()` was deprecated in readr 2.0.0.
i Please use `read_table()` instead.
```

```
Warning in left_join(f_MCSim_read_sim(here::here(file_path, "sim.out")), : Detected an unexpe
```

```
i Row 1 of `x` matches multiple rows in `y`.
i Row 1 of `y` matches multiple rows in `x`.
i If a many-to-many relationship is expected, set `relationship =
  "many-to-many"` to silence this warning.
```



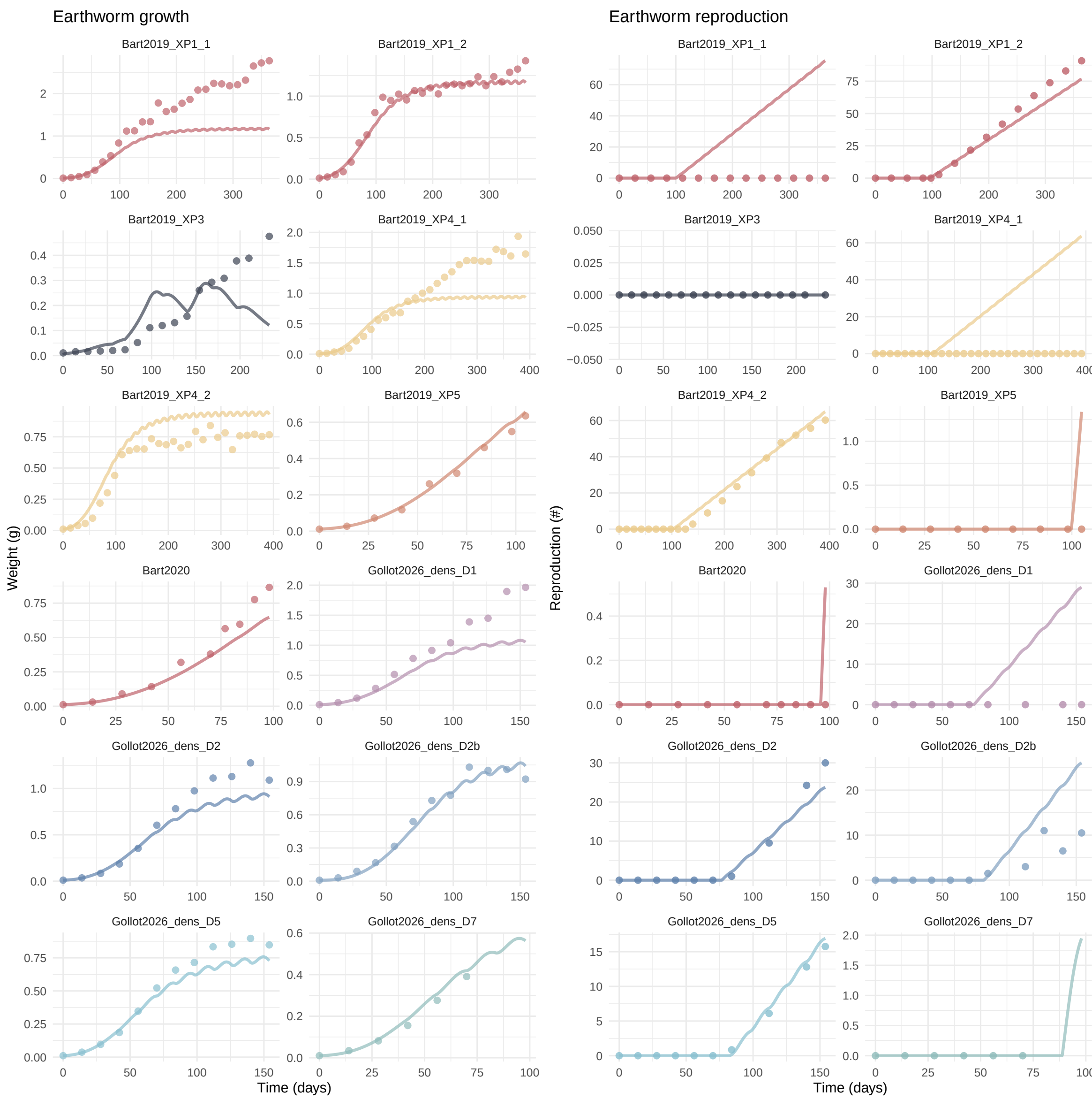
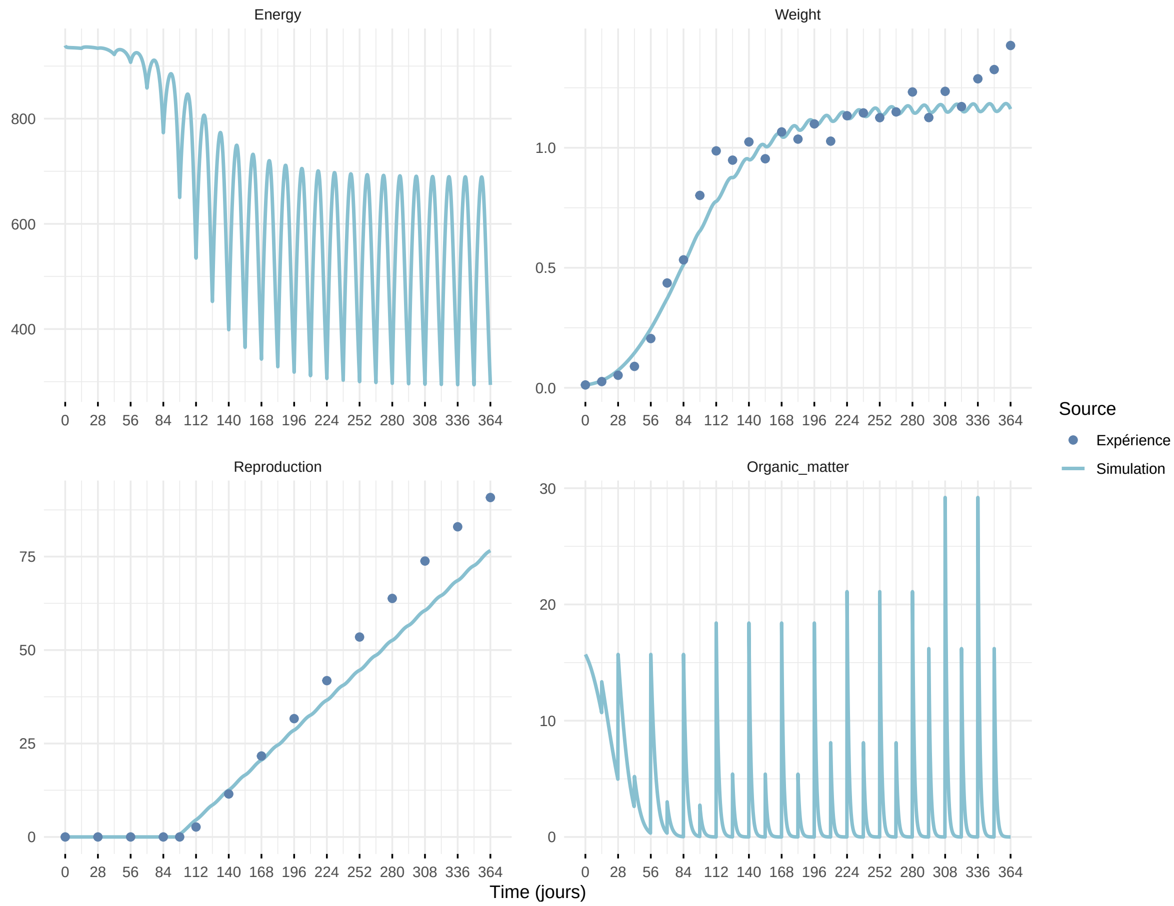
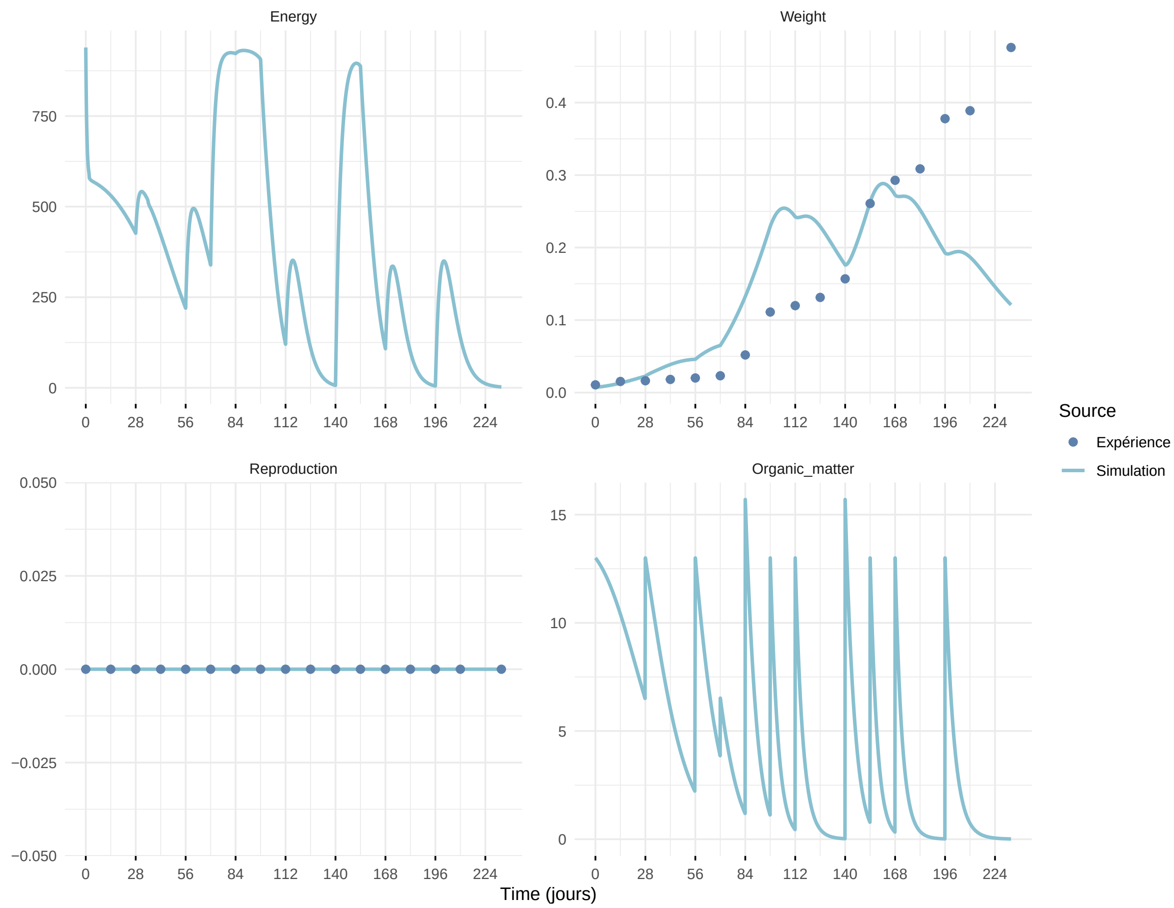


Figure 7: Simulations Weight and Reproduction.

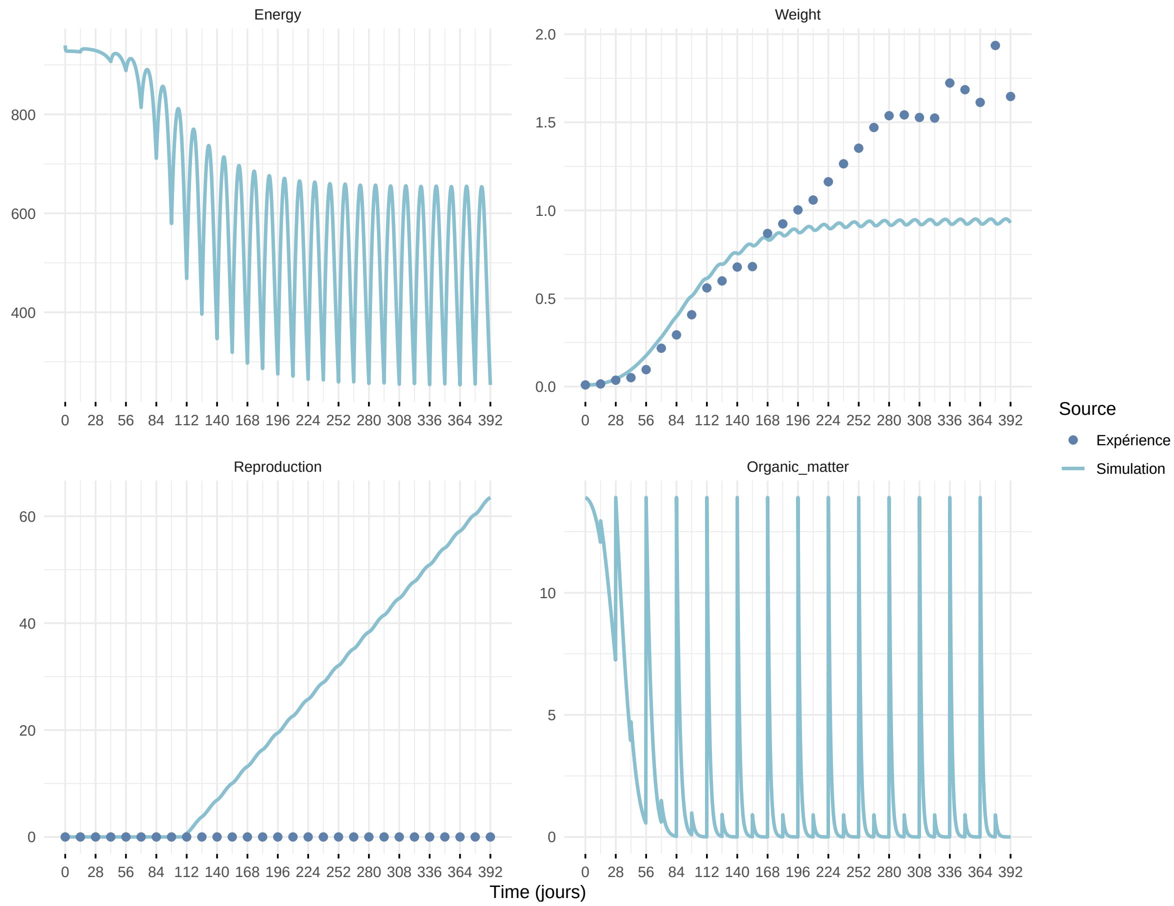
Expérience : Bart2019_XP1_2



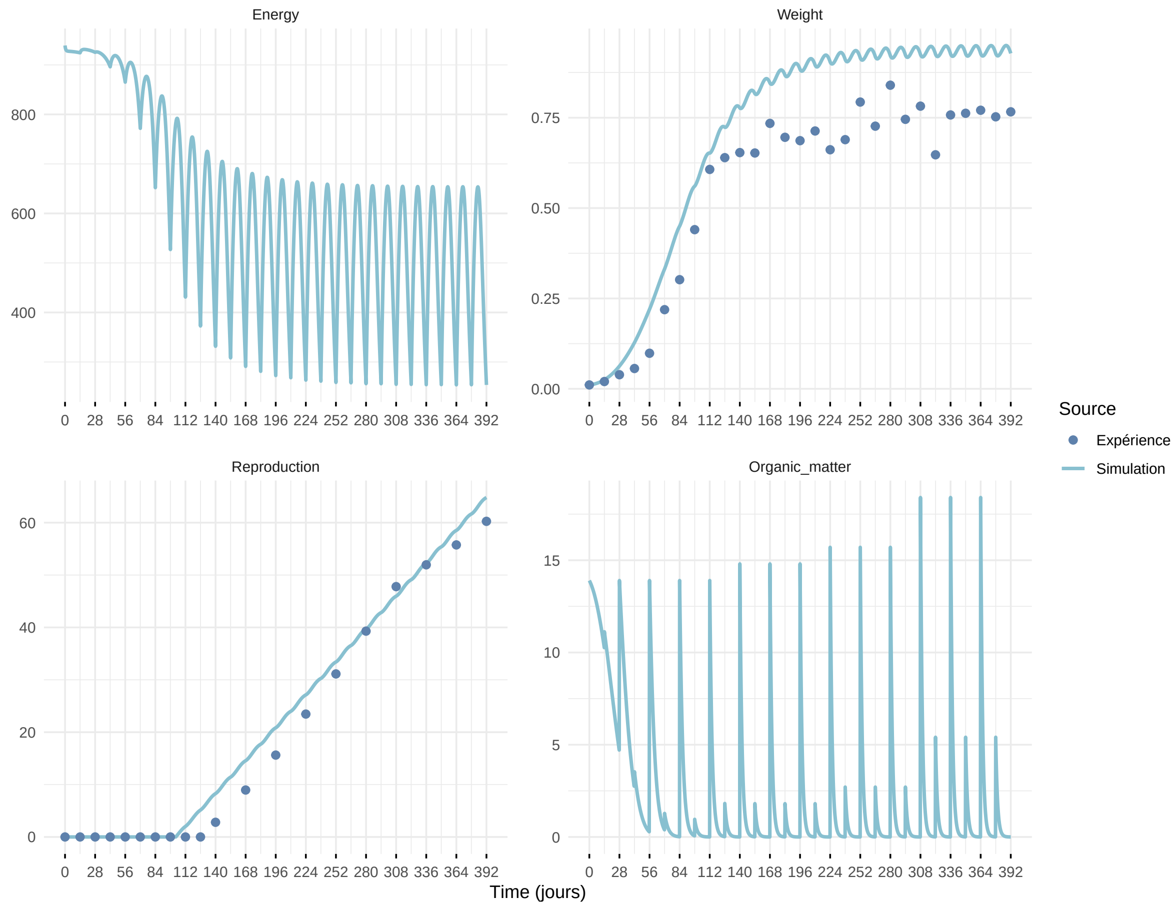
Expérience : Bart2019_XP3

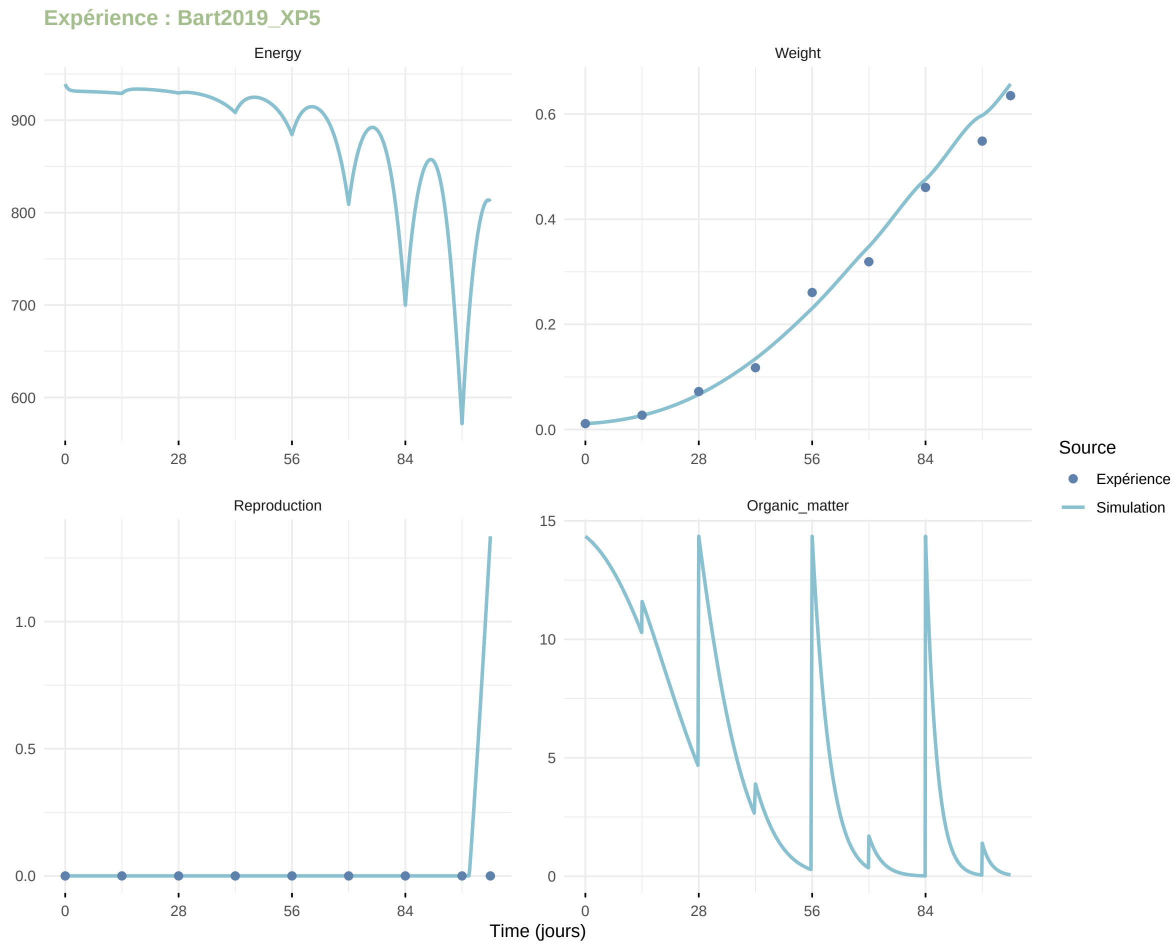


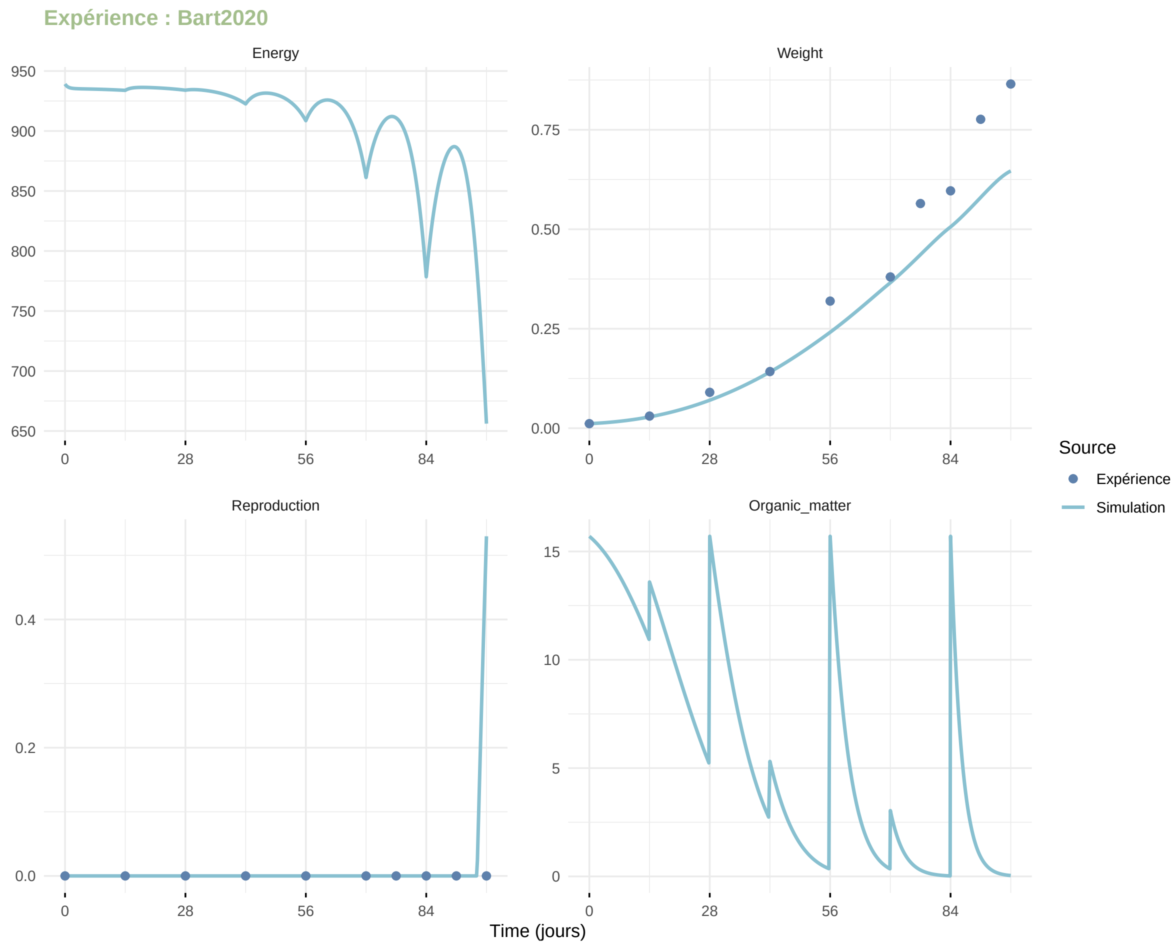
Expérience : Bart2019_XP4_1



Expérience : Bart2019_XP4_2

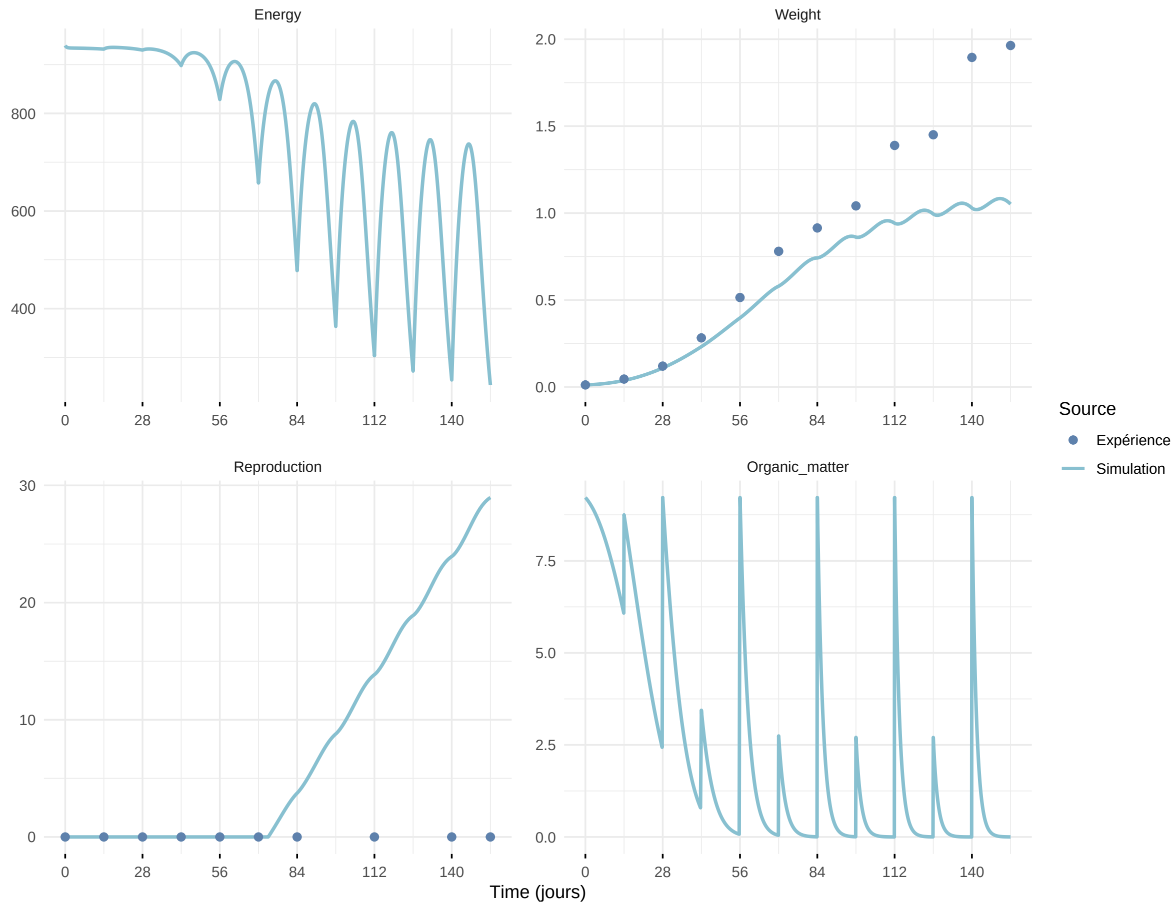


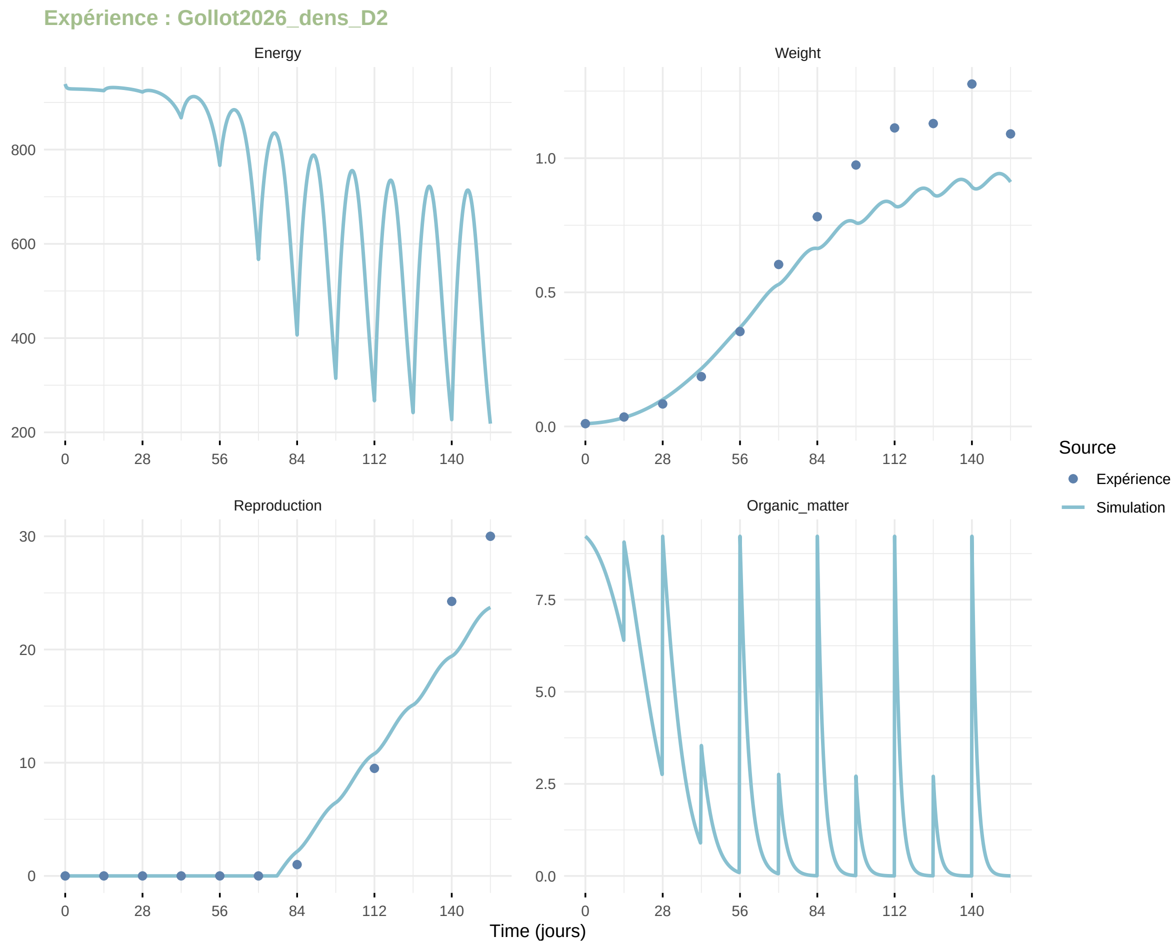

Expérience : Bart2019_XP5
Energy
Weight
Reproduction
Organic_matter
Time (jours)
Source
Expérience
Simulation

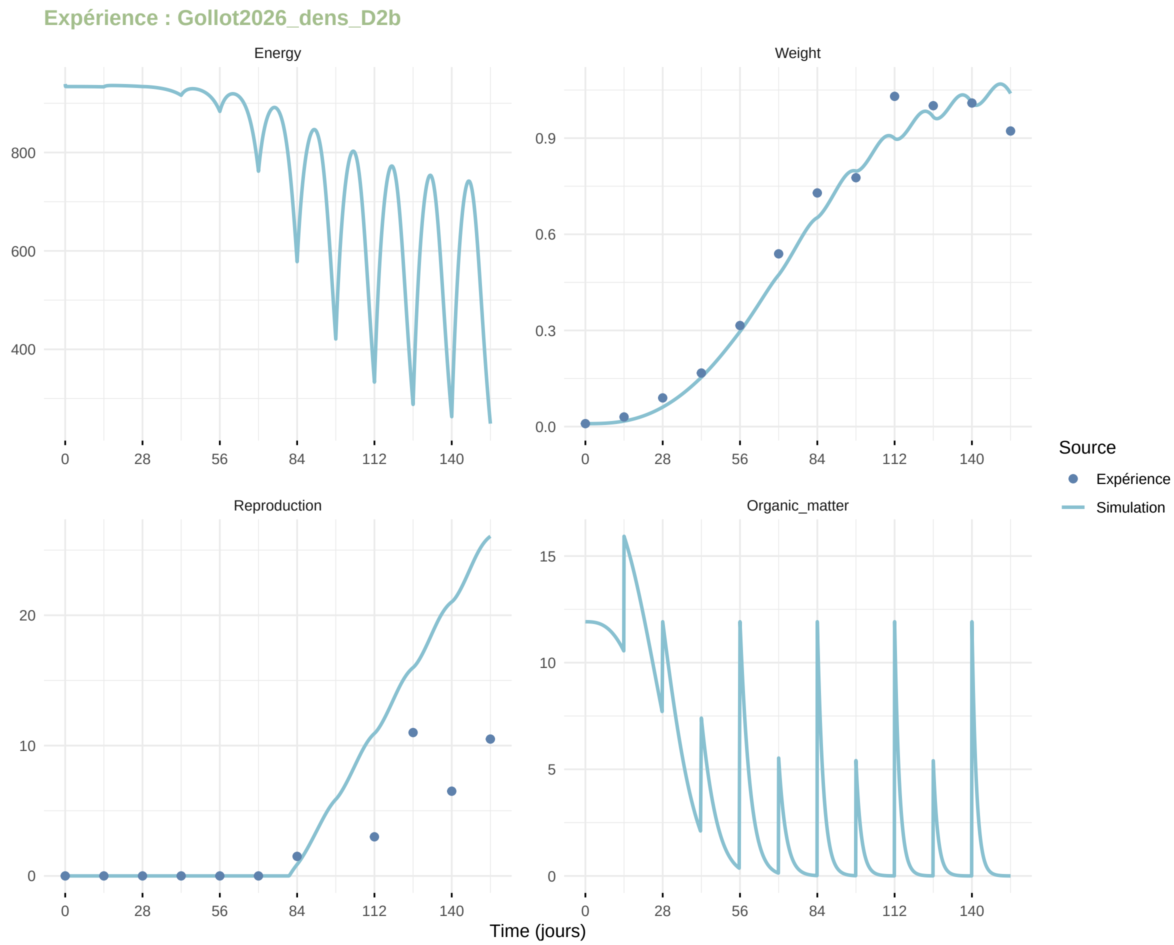


Expérience : Bart2020
Energy
Weight
Reproduction
Organic_matter
Source
Expérience
Simulation
Time (jours)

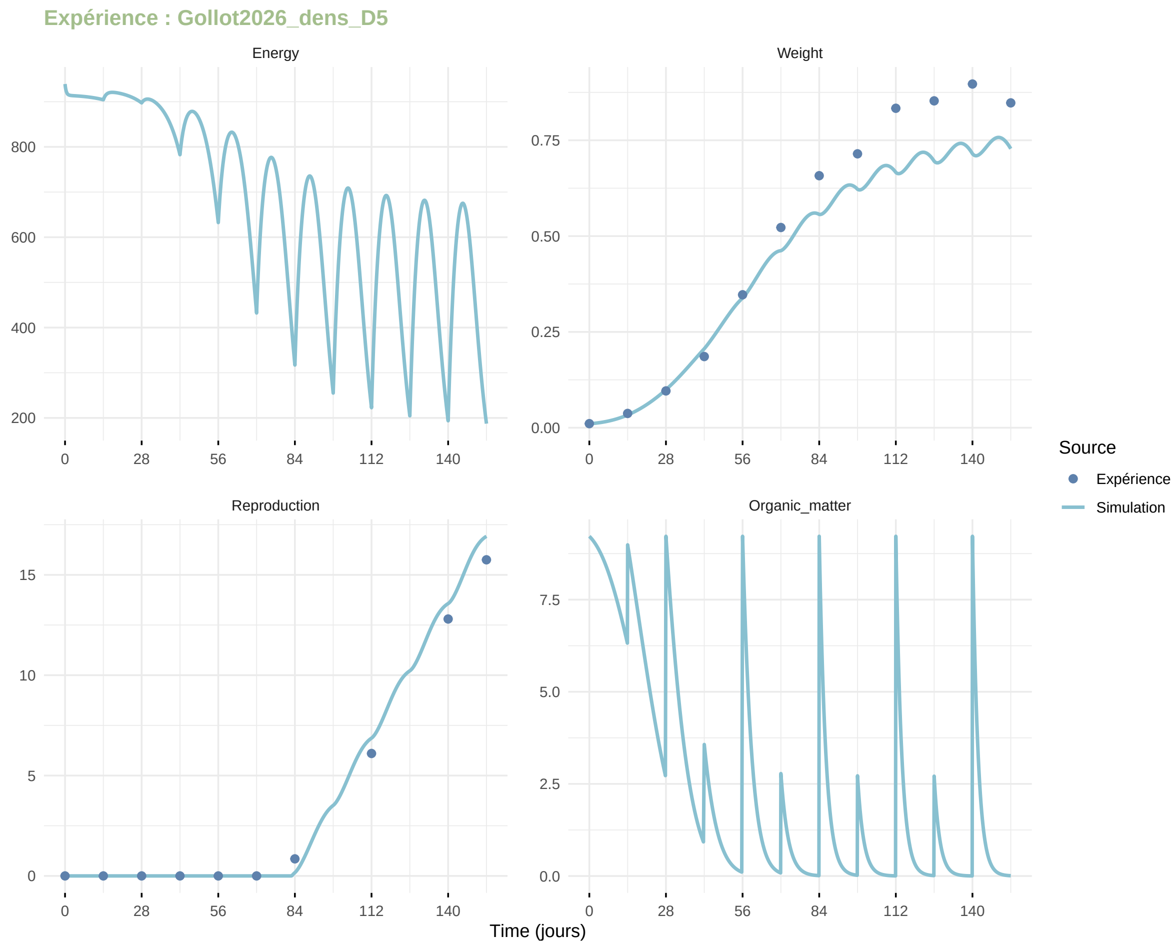
Expérience : Gollot2026_dens_D1




Expérience : Gollot2026_dens_D2
Energy
Weight
Reproduction
Organic_matter
Source
Expérience
Simulation
Time (jours)



Expérience : Gollot2026_dens_D2b
Energy
Weight
Reproduction
Organic_matter
Time (jours)
Source
Expérience
Simulation



Expérience : Gollot2026_dens_D5
Energy
Weight
Reproduction
Organic_matter
Source
Expérience
Simulation
Time (jours)

Expérience : Gollot2026_dens_D7